Remanent Magnetization of the Permian Cutler Formation of Western Colorado[1]

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The early paleomagnetic observations from the Permian period of the western United States were not subjected to demagnetization studies. Observations on the Lower Permian Cutler formation of western Colorado suggest that many of these older poles are in error by 10 or more degrees. The new pole from the Cutler formation is at 42.7°N, 121.9°E ($\alpha_{95} = 6.0$) and is almost identical with the demagnetized results from eastern North America; it provides an excellent test of the dipolar hypothesis. The results of this study emphasize the need for additional study of the older paleomagnetic data for North America.

Recently, *Farrell and May* [1969] have reported on measurements made on samples of the Cutler formation from the Monument Valley region of northern Arizona. Most of their samples proved to be unstable, and only a few of those from the lowest member of the formation, the Halgaito tongue, gave reliable results. Moreover, Farrell and May pointed out that all earlier results [*Graham*, 1955; *Runcorn*, 1955, 1956; *Doell*, 1955; *Collinson and Runcorn*, 1960] from other Permian formations of the Colorado Plateau should be regarded as unreliable, owing to the lack of information regarding stability. *Graham* [1955] reported reconnaissance paleomagnetic observations of samples from the Cutler formation of western Colorado and Utah. The pole positions calculated from these results have been cited extensively as being representative of Permian poles for North America; however, they are likely to be in considerable error, for demagnetization studies were not performed. Thus, it seems essential that rocks of Permian age from the western United States be studied so that a reliable point on the polar wander path for North America can be established.

For such a purpose, two stratigraphic sequences within the Cutler formation in western Colorado were sampled by means of a portable gasoline powered core drill [*Helsley*, 1967*a*]. All orientations were made by means of a magnetic compass corrected for local magnetic variation. Upon return to the laboratory, the 2.54-cm diameter field cores were cut to 2.54-cm long cylindrical samples and were measured on an air-turbine spinner magnetometer [*Helsley*, 1967*b*]. Later measurements made during the demagnetization studies were made on a PAR SM-1 magnetometer.

Regional Geology and Age

The samples used in this study were taken from the lower portion of the Cutler formation exposed in the Gateway quadrangle of western Colorado [*Cater*, 1955] (38°40′N, 109°57′W). Within this region the Cutler formation thins abruptly against the Uncompahgre Uplift, which seems to have been the source for much of the material in the formation. The Cutler formation unconformably overlies the Hermosa formation of Pennsylvanian age and is unconformably overlain in this region by the Lower Triassic Moenkopi formation [*Shoemaker and Newman*, 1959]. Based on regional correlations, this portion of the Cutler is equivalent to, or lies beneath, the Organ Rock sandstone and may be in part equivalent to the Cedar Mesa sandstone member or the Halgaito tongue of the Cutler in the Monument Valley region [*Stewart*, 1959]. Thus, although the age of the formation cannot be stated precisely, owing to lack of fossil control, it is certainly Lower Permian and most likely lowest Permian.

Within the sampling region, the formation

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consists primarily of poorly sorted maroon to purple coarse sandstone with local conglomerate units. Bedding is generally massive and poorly developed. Local finer grained units consisting of poorly sorted, red-brown, medium to fine sandstones and sandy mudstone units are interbedded within the coarse clastic sequence. Three units consisting of these finer grained lithologies were sampled from two different areas. Each of the units covered 10 to 20 meters of section, with core samples up to 12 cm long being taken at approximately one meter stratigraphic intervals. These three fine grained units have a total stratigraphic range of about 200 meters.

The regional strike of the Cutler in this area is uniformly to the northwest, and the local dip varies between 4 and 5 degrees to the southwest. Some of the dip may be due to a sloping surface at the time of deposition. However, the dip of the bedding in the siltstone units is almost identical with that in the conglomerate and sandstone units, and thus it is thought that most of the dip was acquired structurally prior to the deposition of the overlying Moenkopi formation.

DISCUSSION OF RESULTS

The natural remanent magnetization (NRM) and thermal demagnetization results are summarized in Figure 1 and Table 1. The NRM directions are initially quite scattered and have a tendency to streak toward the present earth's field. Most of the measurements lie in the SE quadrant in the lower hemisphere, although there are a few samples that have shallow negative inclinations to the SE. These few samples in the upper hemisphere are the only samples that show NRM directions which do not change on demagnetization.

Remeasurement of the NRM of a portion of the specimens over a year after initial measurement indicated that many of the samples have large viscous moments. Alternating current demagnetization techniques removed this viscous moment but were unsuccessful in removing all the secondary magnetization, for, although a better grouping resulted, the directions of magnetization still showed evidence of a streak toward the present field direction. Progressive thermal demagnetization experiments, however, resulted in a marked decrease in scatter, and thus heating experiments were performed on all samples. The results of these progressive heating experiments are shown in Figure 1 for 300°, 400°, 500°, and 600°C.

Most of the change in direction of magnetization occurred at temperatures below 300°C, since the samples from both localities group well at both 300° and 400°C and have approximately the same mean directions. This would suggest that all the secondary component has been removed. Demagnetization at 500°C continued to move the mean direction away from the present field direction, although the results are much more scattered. This increased scatter may be due to an error in the demagnetization technique, for a small field was noted during the cooling of some of the samples. After correction of this error, a group of the suspect samples was reheated to 600°C. The directions for these samples after cooling were almost all in the upper hemisphere. Consequently, all the remaining samples also were heated to 600°C. The resulting directions are shown in the right-hand plots in Figure 1. The directions from both localities are well grouped and have approximately the same dispersions (see Table 1) as do the clusters for 300° and 400°C; nevertheless the mean directions of magnetization are about 15° from those for the lower temperatures and are farther away from the present field direction.

It is thought that the results for 600°C represent the primary magnetization direction, for they are not significantly different from the mean direction calculated for 500°C after removal of the three widely scattered samples (see Table 1, site 2). Since some of the samples were known to have been cooled in a small applied field, this selection seems justified. Several samples from site 2 scattered widely after heating to 600°C and probably no longer provide reliable estimates of the Permian field direction. Four of these samples have been omitted in the calculation of the final mean direction and pole position (the omitted samples are marked by a minus sign in Figure 1).

POLE POSITIONS

Most of the paleomagnetic studies made on Permian rocks from the western United States have not applied demagnetization studies to test for stability. The results of this study and those of *Farrell and May* [1969] show that this

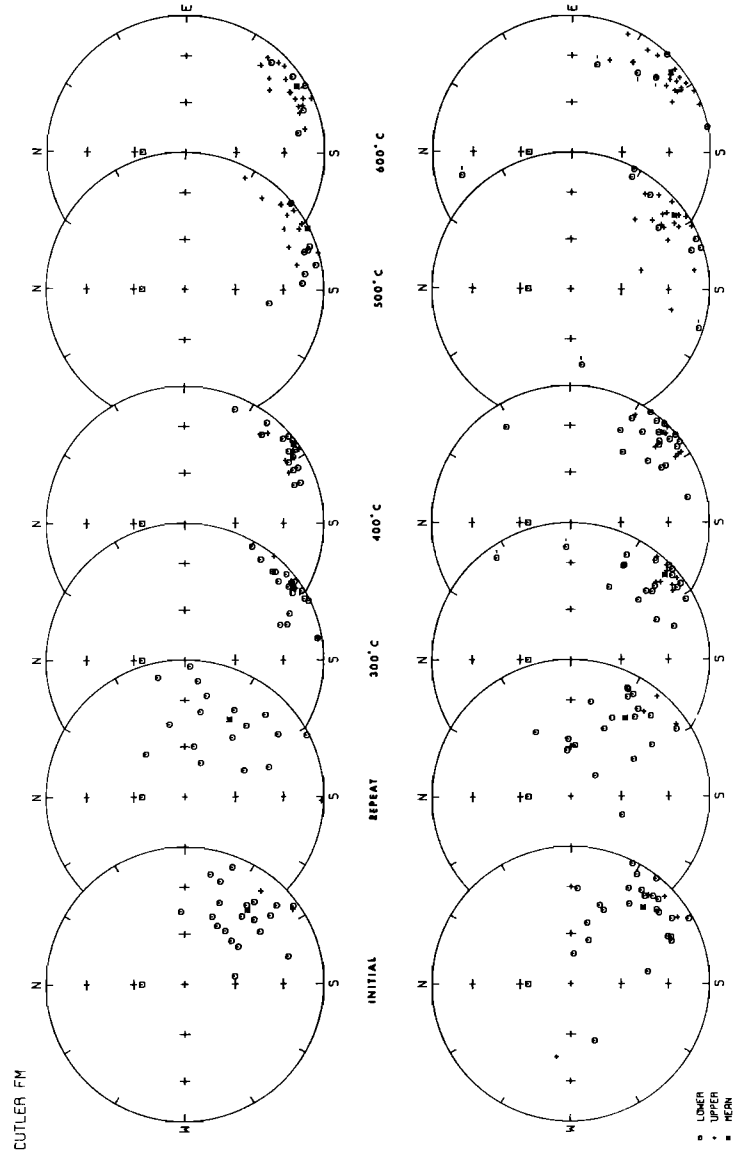


Fig. 1. NRM of Cutler formation before and after progressive heating experiments: site 1, upper half; site 2, lower half. The minus signs indicate samples not used in the statistical calculation.

may be a serious shortcoming. The pole positions derived from the 600°C thermally demagnetized data (see Table 1) are in excellent agreement with earlier published data from Prince Edward Island, West Virginia, and portions of the Colorado Plateau. However, some of the Colorado Plateau data have poles that are quite far removed from these consistent poles (see Table 2). Most of these divergent poles are from early studies in which no demagnetization studies were made. Thus it seems reasonable to try to make some form of reliability assessment in order to use only the data that provide a reliable estimate of the pole position.

The reliability assessment given in the last column of Table 2 is based on the internal grouping of the data and on whether demagnetization experiments have been performed. Obviously, this is a somewhat subjective judgement; nevertheless, all those marked 'good' have similar pole positions, whereas those marked 'fair' or 'poor' generally have pole positions in the vicinity of those found in the course of this study prior to or during demagnetization, thus suggesting that secondary components may be present. The data of Farrell and May for the Toroweap and Hermit show similar disagreements; thus, they may also contain a secondary component.

If one examines the 'good' data from Table 2, all of which are of Upper Carboniferous or Lower Permian age except for Farrell and May's Hoskinnini, which may be as young as Lower Triassic, then one finds a remarkable similarity in pole position. The mean pole position obtained by giving each 'good' Lower Permian result equal weight is 42.3°N, 123.5°E and has an α_{95} of 3.1°. Since these data come from three widely separated regions of North America, their excellent agreement provides a very good confirmation of the dipolar nature of the field for Lower Permian time. Even if the 'fair' data are included, the agreement is still very good (average pole position 41.6°N, 121.5°E and α_{95} of 4.7°).

The largest group of data judged to be of 'poor' reliability is that of *Graham* [1955]. No demagnetization experiments were performed on these samples, and Graham's primary purpose was not to determine precise pole positions but instead to show that indeed the pole position had been different in earlier times. Three of Graham's results give pole positions that are in good agreement with the demagnetized results of later investigators. Nevertheless, it is felt that one should accept Graham's view (J. W. Graham, personal communication, 1968) that his early non-demagnetized data should no longer be used for precise pole position determinations or comparisons. A similar statement

TABLE 1. Results of Analysis of Cutler Formation Data

Site Temp °C	Dec	Inc	N	R	K	α_{95}	Pole long	Pole lat	dp	dm
1	130.1	30.4	22	19.99	10.4	10.1	121.5	17.9	6.2	11.2
1 *	119.8	35.1	18	15.02	5.7	15.9	128.1	9.2	10.6	18.4
1 300	146.0	7.0	21	20.27	27.4	6.2	115.8	37.4	3.1	6.2
1 400	143.8	2.7	21	20.22	25.5	6.4	119.6	38.0	3.2	6.4
1 500	152.8	−1.3	21	19.61	14.4	8.7	111.0	44.6	4.3	8.7
1 600	148.0	−7.4	21	20.21	25.2	6.5	119.0	44.6	3.3	6.5
2	133.2	24.8	28	24.29	7.3	10.9	121.3	22.4	6.3	11.7
2 *	124.5	31.2	22	18.87	6.7	12.9	125.5	14.1	8.1	14.5
2 300	133.8	9.8	27	24.50	10.4	9.1	126.4	29.0	4.6	9.2
2 300 †	137.9	9.0	25	23.65	17.7	7.1	123.0	31.9	3.6	7.1
2 400	135.6	7.5	27	25.10	13.7	7.8	125.7	31.0	4.0	7.9
2 500	145.9	−10.5	27	24.31	9.7	9.4	122.7	44.6	4.8	9.6
2 500 ‡	143.9	−8.5	24	22.60	16.4	7.5	124.0	42.6	3.8	7.6
2 600	140.1	−6.4	27	25.35	15.7	7.2	127.0	39.3	3.7	7.3
2 600 §	142.4	−6.6	23	22.15	26.0	6.0	124.8	40.9	3.1	6.1

* Remeasured.
† 2 omitted.
‡ 3 omitted.
§ 4 omitted.

TABLE 2. North American Permian Pole Determinations

Sample	Age	Dec	Inc	Sites, no	K	Alpha	Long, E	Lat, N,	dp	dm	Reliability
W. Virginia [*Helsley*, 1965]											
Dunkard series	LP	163.5	7.7	9	176.8	3.9	122.3	44.1	2.0	3.9	Good
Colorado [*McMahon and Strangway*, 1968]											
L Maroon fm	P	134.0	−2.0	8	30.0	10.0	133.0	33.0			Fair
Maroon fm	P	132.0	4.0	22	17.0	8.0	131.0	30.0			Fair
Utah-Arizona [*Farrell and May*, 1969]											
Halgaito		151.0	−11.0		54.0	6.0	117.0	49.0			Good
Hoskinnini		150.0	−16.0		23.0	7.0	121.0	50.0			Good
Toroweap		157.0	3.0		36.0	7.0	103.0	47.0			Fair
Hermit		161.0	−7.0		76.0	8.0	101.0	53.0			Fair
Colorado, Utah, New Mexico [*Graham*, 1955, in *Irving*, 1964]											
Abo fm, New Mex.	LP	160.0	55.0	1	7.0	12.0	88.0	17.0	12.0	17.0	Poor
Abo fm, New Mex.	LP	149.0	8.0	1	5.0	18.0	117.0	42.0	9.0	18.0	Poor
Cutler fm, Colo.	P	140.0	6.0	1			123.0	34.0			Poor
Cutler fm, Utah	P	161.0	33.0	1	96.0	10.0	92.0	33.0			Poor
Sangre de Cristo	CP	175.0	31.0	1	9.0	11.0	81.0	38.0	7.0	11.0	Poor
Yeso fm, New Mex.	LMP	143.0	−1.0	1	99.0	3.0	127.0	41.0	2.0	3.0	Fair
Arizona*											
Supai fm	CP	150.0	11.0	7	35.0	10.0	110.0	40.0	9.0	9.0	Poor
Canada, Prince Edward I. [*Black*, 1964]											
Red beds E Prince Edward I.	CP	176.8	5.5		12.5	5.5	121.7	40.9	2.7	5.5	Good
Red beds Cen Prince Edward I.	CP	170.9	7.2		10.6	5.5	128.3	39.5	2.8	5.5	Good
Red beds NW Prince Edward I.	CP	170.0	5.3		22.6	6.1	129.0	39.8	3.1	6.1	Good
Canada, Prince Edward I. [*Roy*, 1963 in *Black*, 1964]											
Red beds	CP	174.7	8.1	17		7.0	123.2	39.0	3.7	7.4	Good
Colorado [*Helsley*, this paper]											
Cutler fm, site 1	LP	148.0	−7.4	1	25.2	6.5	119.0	44.6	3.3	6.5	Good
Cutler fm, site 2	LP	142.4	−6.6	1	26.0	6.1	124.8	40.9	3.1	6.1	Good

* A combined result: Runcorn, 1955, 1956; Doell, 1955; and Graham, 1955 in *Irving* [1964, 7.51].

can be made regarding the data from the Supai formation [*Runcorn*, 1955; *Doell*, 1955; *Graham*, 1955].

The only other Permian data for North America are those of *McMahon and Strangway* [1968]. Their study was done primarily to define the boundaries of the Kaiman reversed interval, and few demagnetization experiments were made. Samples from the lower portion of the Maroon formation are in good agreement with the demagnetized data from other localities in North America, whereas those from the upper portions of the sampled sequence yield pole positions that are not in agreement with other Permian results. Despite the apparent agreement for samples from the lower Maroon formation, these samples have not been included in the average pole determination, since they apparently have not been subjected to adequate demagnetization studies.

Polar Wandering during the Permian

By using the data of *Farrell and May* [1969], *Helsley* [1965, 1969], and *Black* [1964], an estimate for the amount of polar wandering relative to North America for the Permian can be made. The oldest data come from the Carboniferous to Permian red beds on Prince Edward Island [*Black*, 1964], and these poles are very similar to those of the Cutler found in this study (see Figure 2). The pole positions from the Lower Permian Dunkard series of West Virginia [*Helsley*, 1965] are very similar to these but are displaced a few degrees toward the Lower Triassic poles from the Moenkopi formation [*Helsley*, 1969]. The pole from the Halgaito tongue of the Cutler formation [*Farrell and May*, 1969] is even farther displaced toward the Lower Triassic pole. The somewhat uncertain poles from the younger Permian units, the Hermit shale and Toroweap formation, are similar to the Triassic position but are displaced to the south and east, as would be expected if some secondary component of recent origin had not been removed. (Since these are reversely magnetized units for which the south rather than the north pole is plotted, the effect of adding a secondary component is to move the pole away from rather than toward the present pole. The demagnetization results shown in Figure 1 and Table 1 illustrate this point very well.)

Helsley [1969] has argued that the results from the lowest portion of the Moenkopi formation may well be representative of the uppermost Permian. The pole for this interval (labeled R1 in Figure 2) is slightly displaced toward the Lower Permian position from the rest of the Lower Triassic Moenkopi formation results (labeled M). These results are then indicative of a systematic polar wandering of about 20° during the Permian period. This conclusion should be regarded as suggestive until additional data from rocks of Middle and Upper Permian age become available. It does seem certain however, that approximately 20° of polar movement (continental drift?) occurred between Lower Permian and Lower Triassic time.

Age of the Hoskinnini Tongue

Farrell and May [1969] have described observations made on the Hoskinnini tongue of the Cutler formation in the Monument Valley region of the Colorado Plateau and have found results very similar to those of the Halgaito tongue of the Cutler formation. At first, this seems reasonable. However, *Stewart* [1959] has demonstrated that the Hoskinnini can be shown to be physically equivalent to the lowest member of the Lower Triassic Moenkopi formation,

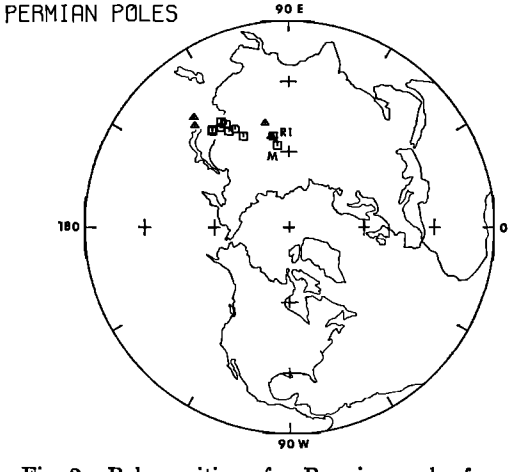


Fig. 2. Pole positions for Permian rocks from North America: octagon, data from Cutler formation; square, published data considered 'good' (see Table 2); triangle, published data considered 'fair.' R and M are poles from the Lower Triassic Moenkopi formation [*Helsley*, 1969].

whereas the Halgaito tongue of the Cutler is the lowest member of the Permian sequence in the Monument Valley region. Moreover, the pole for the Hoskinnini found by Farrell and May is not at all similar to that for the lower Moenkopi found by *Helsley* [1969].

Assuming that Farrell and May did not make a mistake in identifying the Hoskinnini, this indicates that the Hoskinnini cannot be Triassic in age in the Monument Valley region and thus suggests that it may represent a facies that is markedly time transgressive. Until this difficulty in age assignment of this unit is resolved, Farrell and May's Hoskinnini result, despite its good appearance, should not be considered as reliable pole position for either the Permian or the Lower Triassic.

CONCLUSIONS

The results from this study support those of *Farrell and May* [1969], namely that most of the older Permian data from the western United States should not be regarded as reliable indicators of pole positions. Comparison of the pole positions found in this study, in which demagnetization was done, with those from the eastern part of North America provides more convincing evidence for the dipolar nature of the Permian field. It is suggested that more data are needed for rocks of Middle and Upper Permian age and that these data may be useful in solving some of the stratigraphic problems present in the Colorado Plateau region.

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